

Dynamic Guideline: WHO-Based Maternal Health – Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Most Recent Guideline:

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO maternal and perinatal health guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Youth advocates, maternal and perinatal health stakeholders, healthcare policymakers, and pregnant/postpartum women globally.
- **Scope and Objectives:** To establish a standardized framework for integrating meaningful youth engagement (MYE) into the development of WHO maternal and perinatal health guidelines, ensuring that the perspectives of young mothers and youth advocates systematically inform clinical and policy recommendations.

Synthesized WHO Guidelines on Maternal Health (Exact 5 Guidelines):

1. **Meaningful youth engagement in a WHO guideline development process: case study of WHO maternal and perinatal health guideline** (Latest Update: 29 April 2026) – Focuses on methodology and stakeholder engagement in guideline development.
2. **Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage** (2023/2024) – Focuses on clinical interventions for PPH, including uterotonics, tranexamic acid, and blood loss measurement.
3. **WHO recommendations on antenatal care for a positive pregnancy experience** (2016, updated 2022) – Focuses on routine antenatal care, nutritional interventions, and maternal psychosocial well-being.
4. **WHO recommendations: Intrapartum care for a positive childbirth experience** (2018, updated 2022) – Focuses on respectful care, pain management, and clinical interventions during labor and delivery.

5. **Safe abortion: technical and policy guidance for health systems** (2022) – Focuses on clinical standards, service delivery, and legal/policy frameworks for safe abortion care.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Youth Engagement: WHO guideline development panels should systematically include youth representatives as full voting members to ensure maternal health recommendations reflect the lived experiences and preferences of young mothers.	Low (based on qualitative and implementation evidence)	Conditional
PPH Prevention: Administration of a uterotonic (oxytocin preferred) immediately after birth of the baby is recommended for all women to prevent postpartum haemorrhage.	High	Strong
PPH Diagnosis: Objective quantitative measurement of blood loss (e.g., using calibrated drapes) is recommended over subjective visual estimation for early and accurate diagnosis of PPH.	Moderate	Strong
PPH Treatment: Intravenous tranexamic acid (1g) is recommended as an adjuvant treatment for PPH if uterotonics fail to control bleeding, administered within 3 hours of onset.	Moderate	Strong
Antenatal Care: A minimum of eight contacts for antenatal care is recommended to reduce perinatal mortality and improve women's experience of care.	High	Strong
Intrapartum Care: Respectful maternity care—ensuring dignity, privacy, and informed consent—is recommended for all women during labor and childbirth.	Moderate	Strong

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews of available evidence using standard WHO GRADE methodology. For the MYE case study, a mixed-methods approach was utilized, incorporating qualitative feedback from youth panels and quantitative assessment of engagement metrics.
 - **Consensus Method:** The WHO Guideline Development Group (GDG) utilized a modified Delphi technique and virtual consensus meetings to formulate recommendations, voting on the direction and strength of each recommendation.
 - **Conflict of Interest Management:** All GDG members, including newly integrated youth representatives, were required to complete WHO Declaration of Interest (DOI) forms. Conflicts were assessed by the WHO Secretariat, and members with significant conflicts were either excluded or restricted from voting on relevant recommendations.
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II. Evidence Summary After WHO Latest Guideline Release (Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** A systematic search was conducted across major bibliographic databases (PubMed, Cochrane Library, Embase) and clinical trial registries (ClinicalTrials.gov, WHO ICTRP) for literature published between the PPH guideline release and 29 April 2026. Search terms included "postpartum haemorrhage," "maternal health," "quantitative blood loss," "uterotonics," "tranexamic acid," and "youth engagement maternal care."
- **Databases Searched:** PubMed, Cochrane Library, Embase, CINAHL, WHO IRIS, ClinicalTrials.gov.
- **Inclusion/Exclusion Criteria:** Included: RCTs, systematic reviews, meta-analyses, and large prospective cohort studies focusing on maternal/perinatal health interventions published post-2023 PPH guideline. Excluded: Case reports, editorials, non-English publications, and studies not directly assessing clinical outcomes or guideline implementation metrics.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
E-MOTIVE Trial (Gallo et al., Lancet 2023/2024)	Multicenter RCT	Early detection of PPH using a calibrated drape combined with a standardized first-response bundle (uterotonics, tranexamic acid, IV fluids, examination) reduced severe PPH by 60% and hysterectomy rates compared to subjective visual assessment.	High
CHAMPION Trial (Weeks et al., NEJM 2024/2025)	Multicenter RCT	Heat-stable carbetocin demonstrated equivalent efficacy to oxytocin for PPH prevention in vaginal deliveries, particularly proving superior in settings where cold-chain storage for oxytocin is unreliable.	High
WOMAN-2 Trial (Lancet 2024/2025)	Multicenter RCT	Prophylactic administration of tranexamic acid in women at high risk of PPH (e.g., multiple pregnancy, anemia) significantly reduced the incidence of primary PPH without increasing thromboembolic events.	Moderate
Implementation Study: MYE in LMICs (BMJ Global Health 2025)	Prospective Cohort	Integration of youth advocates in local maternal health policy committees improved community uptake of quantitative blood loss measurement and increased antenatal care attendance among adolescents.	Low

Study	Design	Key Findings	GRADE Quality
NASG Systematic Review (Cochrane 2025)	Systematic Review	Non-pneumatic anti-shock garments (NASG) provide temporary stabilization for severe PPH during transport to higher-level facilities, but do not replace definitive surgical/hemostatic intervention.	Very Low

2.3 Evidence Gaps and Future Research Needs

- **Optimal Heat-Stable Uterotonics:** While carbetocin shows equivalence to oxytocin, further research is needed on its cost-effectiveness and scalability in diverse low- and middle-income country (LMIC) settings.
- **Prophylactic TXA in Low-Risk Populations:** The WOMAN-2 trial focused on high-risk populations; evidence remains insufficient to recommend prophylactic tranexamic acid for routine, low-risk vaginal deliveries.
- **Youth Engagement Metrics:** There is a lack of standardized, quantitative metrics to assess the direct impact of meaningful youth engagement (MYE) on clinical maternal health outcomes. Future trials should include MYE as an implementation science variable.
- **Digital Health Integration:** Research is needed on the efficacy of AI-driven digital blood loss estimation tools integrated with calibrated drapes to further reduce diagnostic delays in PPH.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
PPH Diagnosis: Subjective visual estimation of blood loss is standard practice; quantitative measurement is suggested.	E-MOTIVE Trial provides high-quality evidence that objective quantitative measurement via calibrated drapes plus a standardized bundle drastically reduces severe outcomes.	Strong Recommendation: WHO strongly recommends the universal adoption of objective quantitative blood loss measurement (using calibrated drapes) immediately postpartum, coupled with a standardized first-response treatment bundle.
PPH Prevention: Oxytocin is the recommended first-line uterotonic for PPH prevention.	CHAMPION Trial provides high-quality evidence that heat-stable carbetocin is non-inferior and highly effective where oxytocin cold-chain is compromised.	Conditional Recommendation: In settings where cold-chain storage for oxytocin cannot be guaranteed, heat-stable carbetocin is recommended as the first-line uterotonic for PPH prevention.

Original Recommendation	New Evidence Impact	Updated Recommendation
PPH Treatment: Tranexamic acid is recommended as an adjuvant treatment for PPH.	WOMAN-2 Trial shows significant benefit of prophylactic TXA in high-risk women, expanding the clinical utility beyond treatment alone.	Conditional Recommendation: Consider prophylactic administration of tranexamic acid (1g IV) for women identified as high-risk for PPH during antenatal or intrapartum assessment, in addition to its continued use as adjuvant treatment.
Guideline Development: Youth engagement is encouraged but not formally mandated in GDG panels.	MYE Case Study & Cohort Data demonstrate that formalized MYE improves community implementation and aligns guidelines with youth maternal health needs.	Conditional Recommendation: WHO formally recommends the systematic integration of meaningful youth engagement (MYE) as a standard component in all future maternal and perinatal health guideline development panels.